Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	Nameless
Project / Solution Name	SME Manufacturing Quotation Assistant
Name (Leader)	ZHANG XINRUI
Email & Phone (Leader)	AIT2609083@xmu.edu.my
University Affiliation(s)	XIAMEN UNIVERSITY MALAYSIA
Live Prototype URL	Pending: supply the verified published prototype URL.
Code Repository URL (public / access granted to judges)	https://github.com/The-Wildest-Dream/SDC-project

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

The browser accesses a Python Streamlit application with pages for data, inquiries, price analysis, approvals and quotation records. Business services calculate costs, validate inputs and preserve quotation versions in PostgreSQL through SQLAlchemy and psycopg. A separately trained, local scikit-learn Ridge model provides a historical-price reference. It does not make final pricing or credit decisions. Model failure or unsuitable input triggers a fallback to available historical references and cost calculations. Replit hosts the development environment; the published production URL and configuration require confirmation. GitHub stores source and commit history. Chinese, English and Malay translation resources support the interface. No runtime external GenAI API or GPU service was verified.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.10+; exact runtime to confirm	Application logic	PSF
Streamlit; installed version to confirm	Web interface	Apache-2.0
pandas / NumPy; versions to confirm	Data processing	BSD-3-Clause
scikit-learn; version to confirm	Ridge regression / preprocessing	BSD-3-Clause
SQLAlchemy; version to confirm	Database access	MIT
psycopg 3; exact version to confirm	PostgreSQL driver	LGPL-3.0-or-later; verify binary distribution notices

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
None verified in application runtime	Local model inference and database access; team to confirm any later additions	Not applicable

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Development, execution and intended publishing	Account plan, region and live deployment to confirm
PostgreSQL via project configuration	Persistent business records	Actual provider / plan / region to confirm
GitHub	Source code and version history	Repository visibility to confirm

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Historical unit-price model; artifact version in evaluation.json	Team training script; scikit-learn Ridge(alpha=1)	Self-trained on CPU; product-specific models	Team-generated artifact; scikit-learn BSD-3-Clause

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Synthetic manufacturing records	generate_data.py; sample_data/ in repository	Synthetic	Team-generated; final repository licensing to confirm	N — fictional customers
Historical price training and evaluation set	300 synthetic transactions from fixed-seed generator	Synthetic	Team-generated; final repository licensing to confirm	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Replit Agent; platform model/version not confirmed	Code generation, debugging and application changes	Major
ChatGPT; exact model/version not confirmed	Planning, development instructions, translation and documentation drafting	Major

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Development used staged instructions and feature-specific checks. The repository contains unit tests and database/workflow verification scripts, but their presence alone does not establish that all checks passed. The team must confirm which tests and browser workflows it personally executed, record results and review generated code before signing. Model limitations and baseline comparisons are documented. This declaration draft also requires team review.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☐ Yes ☑ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Platform; SDC credit redemption to confirm	Application development and execution
RunPod / other sponsors	No verified usage in reviewed implementation	Confirm actual usage and organiser requirements; do not claim unused resources

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit supports the current development workflow and is required by the supplied competition guidance. The Python application and PostgreSQL design are technically portable to another compatible host. The historical-price model runs on CPU and has no verified dependency on RunPod or a GenAI API. Actual sponsored credits and any additional mandatory sponsor integration must be confirmed before submission.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Demonstration customer, product, inventory and transaction records are synthetic. Uploaded CSV records and saved inquiries, quotation versions, approval records and operation logs are stored in the configured PostgreSQL database. Uploaded files could contain personal data; the demonstration should use synthetic records only. DATABASE_URL is read from the environment rather than embedded in documentation. No external model API data transfer was verified. Role switching is a simulation, not authenticated access control. No automatic retention/deletion policy was verified; records remain until managed by the operator. Production access settings and retention arrangements need team confirmation.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Synthetic-data results demonstrate implementation, not real-world pricing accuracy or commercial benefit. The regression estimates historical prices rather than optimal profit or purchase probability. Documented results show that some product models perform worse than the median baseline. Small samples, unseen products and out-of-range inputs limit applicability. Cost floors, human approval, explicit model fallbacks and version snapshots reduce inappropriate use. Payment-history summaries do not establish honesty or justify automatic credit approval. Rule-based financing costs depend on stated assumptions. Simulated roles are not production authentication. Live deployment, persistence and multilingual behaviour require final end-to-end verification.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- [x] All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- [x] All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- [x] No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- [x] Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- [x] Any use of official SDC sponsor resources has been declared in Section 5.
- [x] The prototype described here is accessible to judges through the links provided in Section 1.
- [x] All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]

[Date]